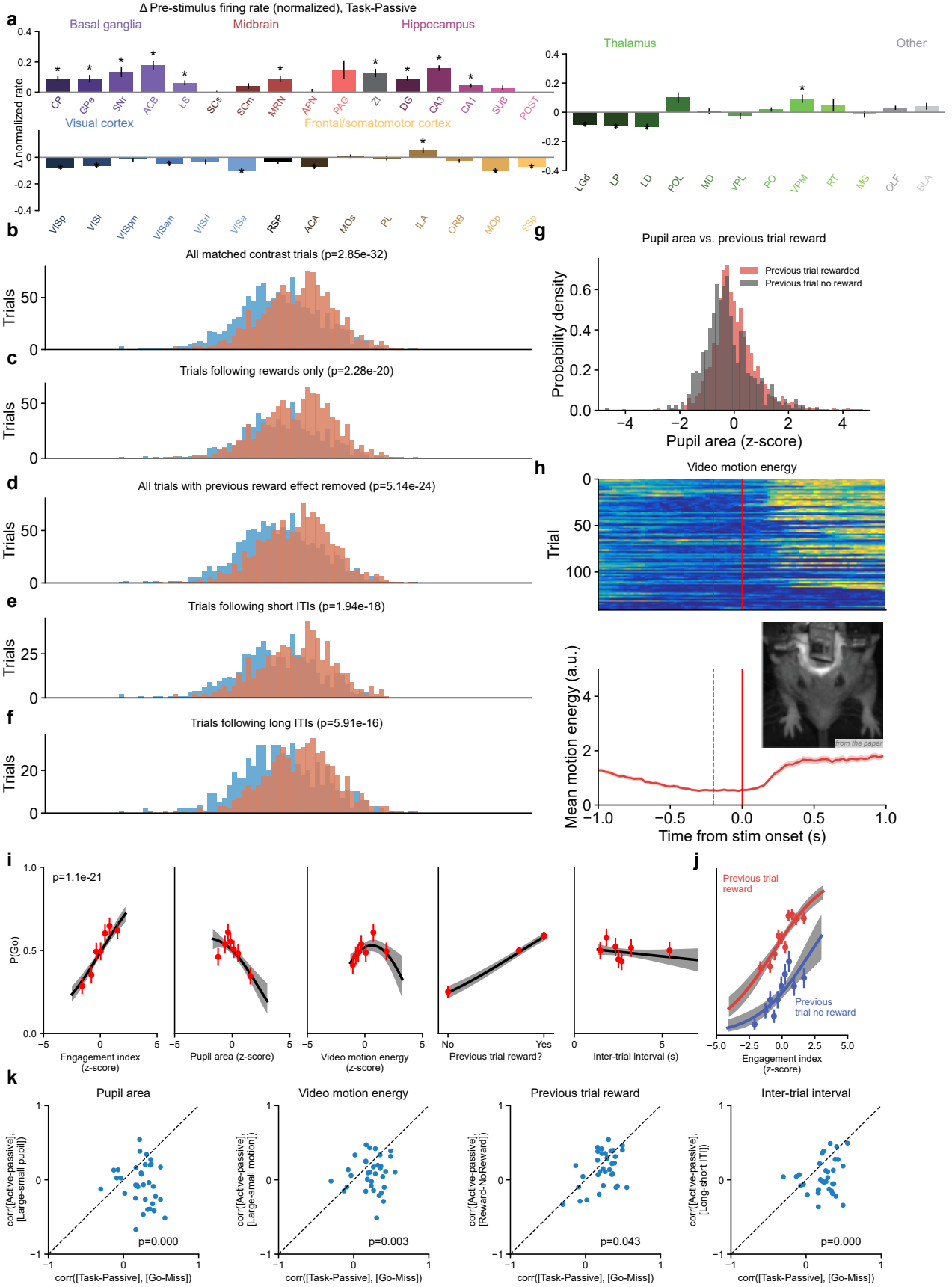

a
Δ Pre-stimulus firing rate (normalized), Task-Passive
Basal ganglia
Midbrain
Hippocampus
Thalamus
Other
Visual cortex
Frontal/somatomotor cortex
Δ normalized rate
CP
GPe
SNr
ACB
LS
SCs
SCm
MRN
APN
PAG
ZI
DG
CA3
CA1
SUB
POST
LGd
LP
LD
POL
MD
VPL
PO
VPM
RT
MG
OLF
BLA
VISp
VISl
VISpm
VISam
VISrl
VISa
RSP
ACA
MOs
PL
ILA
ORB
MOp
SSp
b
All matched contrast trials (p=2.85e-32)
c
Trials following rewards only (p=2.28e-20)
d
All trials with previous reward effect removed (p=5.14e-24)
e
Trials following short ITIs (p=1.94e-18)
f
Trials following long ITIs (p=5.91e-16)
Trials
g
Pupil area vs. previous trial reward
Previous trial rewarded
Previous trial no reward
Probability density
Pupil area (z-score)
h
Video motion energy
Trial
Mean motion energy (a.u.)
Time from stim onset (s)
from the paper
i
p=1.1e-21
P(Go)
Engagement index (z-score)
Pupil area (z-score)
Video motion energy (z-score)
Previous trial reward?
Inter-trial interval (s)
j
Previous trial reward
Previous trial no reward
Engagement index (z-score)
k
Pupil area
Video motion energy
Previous trial reward
Inter-trial interval
corr([Active-passive], [Large-small pupil])
corr([Active-passive], [Large-small motion])
corr([Active-passive], [Reward-NoReward])
corr([Active-passive], [Long-short ITI])
corr([Task-Passive], [Go-Miss])
p=0.000
p=0.003
p=0.043
p=0.000